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STANDARD OPERATING PROCEDURE

TITLE: OPERATIONS, HAZARDOUS MATERIALS
INCIDENT RESPONSE

NUMBER: S 6601 R6

EFFECTIVE: AUGUST 17, 2018

REPLACES: S 6601 R5 (9.19.16)

DATE REVIEWED: SEPTEMBER 19, 2016

Purpose:

To establish procedures for responding to hazardous materials (Hazmat) incidents.

General Information:

- I. All Operations personnel must be competent in performing the procedures established in this publication.
- II. The Fire Department is responsible for managing hazardous materials related incidents in the city of Fort Worth and, by way of approved mutual / automatic aid agreements, in neighboring cities as well. The Fire Department has established, trained and equipped a Hazardous Materials Team (Hazmat Team). The team will be dispatched to all hazardous materials incidents in the city and, when requested and authorized, to similar incidents in the areas surrounding Fort Worth.
- III. The Hazmat Team:
 - A. Composition (Primary Units)
 1. (3) Hazmat Battalion Chiefs
 2. (3) Captains
 3. (6) Lieutenants
 4. (9) Engineers
 5. (18) Firefighters
 - B. Units:
 1. Primary units include B-2, S-2, E-2 and T-2
 2. In addition to the primary units, satellite units will include B-1, B-3, B-4, B-5, B-6, TACMED, and Stations 17, 23, 24, and 37.
 - C. Responsibility:
 1. Battalion Chiefs and station officers are responsible for the facilitation of training, equipment maintenance, personnel management, and for meeting applicable standards of federal and state agencies having jurisdiction.
 2. Coordination of the Hazmat program is the responsibility of the Hazmat Coordinator. Responsibilities of the Hazmat Coordinator are as follows:
 - A. Establishing overall direction and purpose of the Hazmat Program.

- B. Planning, establishing and coordinating annual and long term goals for the Hazmat Program.
- C. Coordination of training schedules through the Master Scheduler with approval from the appropriate Battalion Chief.
- D. Identifying and supporting all training and equipment needs within the Hazmat Program.
- E. Keeping all participants within the Hazmat Program trained, informed and equipped to provide the best service possible to the citizens of Fort Worth and its surrounding communities.

D. Training:

- 1. Both primary and satellite unit members will be trained to the Hazmat Technician level as directed by applicable federal and/or state laws and regulations.
 - 2. Each Hazmat member will attain a minimum of 8 hours of hazmat training to satisfy applicable federal and/or state laws and regulations.
 - 3. Battalion Chiefs will attain the training level of Hazmat Technician and maintain 8 hours per year of approved hazmat related training.
 - 4. Constant staffing will include a minimum of 4 Battalion Chiefs on duty at all times, this includes the Safety Chief.
- IV. To provide additional technical assistance to cope with these situations, the Fire Department will enlist the aid of other City departments as well as outside governmental and private agencies with expertise in handling hazardous materials. Assistance is also available from the Emergency Management Office Technological Hazards Officer.
- V. The term "hazardous materials spill" is used to denote the actual release of a substance (gas, liquid or solid) which threatens the health, safety or welfare of citizens or the destruction of property, either public or private.
- VI. Fixed sites will maintain Material Safety Data Sheets (MSDS) somewhere in the facility for each chemical manufactured or used. It may be necessary to contact a facility employee to retrieve the MSDS and/or to accurately identify the materials involved.
- A. MSDS will normally provide very specific information in the following categories:
- 1. Material identification
 - 2. Ingredients and hazards
 - 3. Physical data
 - 4. Fire and explosion data
 - 5. Reactivity data
 - 6. Health hazard information
 - 7. Spill, leak and disposal procedures
 - 8. Special protection information
 - 9. Special precautions and comments

B. The Joint Emergency Operations Center (JEOC) may have additional information on stored chemicals and their hazards.

VII. Identification of the material(s) involved is critical. Hazardous materials transported by truck or rail will have a placard bearing a United Nations (UN) identification number attached to the sides and/or ends of the container (trailer, tank, or rail car). Aircraft do not have such placards.

VIII. Shipping papers, which specifically identify the material, will normally be found in the truck cab or in the possession of the truck driver, a train crewmember, or aircraft pilot.

IX. All Operations personnel must be familiar with the material identification coding systems of the Department of Transportation (DOT) and the (UN), which is found in the North American Emergency Response Guidebook (ERG).

A. All injuries reported at Hazmat Level I incidents will dictate a response by TACMED and a Battalion Chief in whose territory the incident occurs.

B. Response by B-2 out of their territory will be optional and at their discretion or at the discretion of the Incident Commander.

X. For materials not listed in the ERG, refer to the appropriate MSDS, Shipping Papers, CHEMTREC, or the Emergency Management Office.

XI. Hazmat Incident Safety Zones

A. Hot Zone:

1. The geographic area of an incident for which it is immediately dangerous to life and health (IDLH). The Hot Zone is defined by the following parameters:

- a. Oxygen Deficient; below 19.5%
- b. Oxygen Enriched: above 23.5%
- c. Explosive: 10% Lower Explosive Limit (LEL) or greater
- d. Poison; Referenced values of IDLH or Short Term Exposure Limit (STEL), 10 times Threshold Limit Value – Time Weighted Average (TLV-TWA) is also acceptable, STEL = 3 X TLV-TWA for no more than 30 minutes.
- e. Radioactive: 2mR per hour
- f. Or as determined by the Incident Commander and Hazmat Branch Director

2. Access to the Hot Zone is to be rigidly controlled. Only those persons necessary to control the incident may enter. They will:

- a. Be part of any entry team (no individuals).

- b. Wear appropriate personal protective equipment and chemical protective clothing.
- c. Be assigned a task by the IC and HazMat Branch, through the Entry Group Supervisor.
- d. Trained to the Hazmat Technician Level
- e. Have Decontamination (Decon) established
- f. **ALL OTHER PERSONS MUST STAY OUT — NO EXCEPTIONS!**

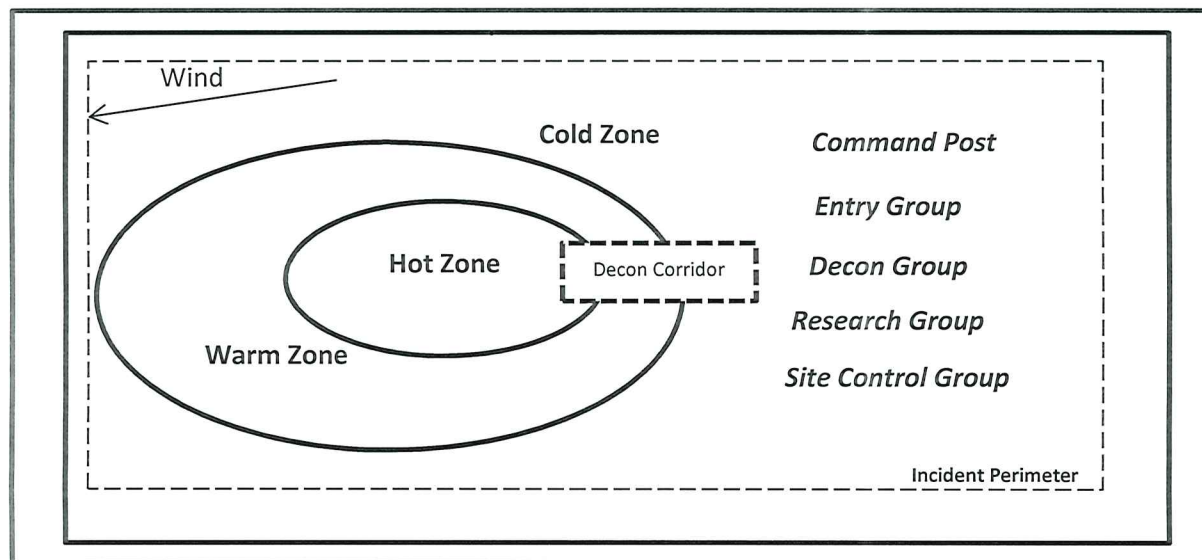
B. Warm Zone:

1. This area is bound by the Hot Zone and the Warm Zone. The Decon Corridor will be located inside this area. The Decon Group will be established in the Cold Zone but deploy into Warm Zone. The Decon Group will be used to control the amount of contamination leaving the Hot Zone.
2. As few personnel as possible should be committed to this zone to reduce confusion. If the situation were to change (i.e., wind change, uncontrolled chemical release, explosion, etc.), tactical adjustments could be accomplished more efficiently.
3. Not all the Decon Group will be utilized at one time within the Decon Corridor (within the Warm Zone). The Decon Group Supervisor, Decon Backup Teams and Decon support personnel will stay in the Cold Zone.

C. Cold Zone:

1. The Cold Zone contains the Command Post, Entry Group, Decon Group, Research Group, Site Control Group and such support functions as are deemed necessary to control the hazardous materials incident.

D. Hazardous Materials Incident Diagram



XII. Training:

- A. As noted previously, personnel from Stations 2, 17, 23, 24, and 37, TACMED, Battalion 1, Battalion 2, Battalion 3, Battalion 4, Battalion 5, Battalion 6, EMS 1, EMS 2, and EMO staff will be trained to the Technician level for Hazmat response.
- B. All other personnel assigned to Operations must be trained to the Operations Level as required by the Texas Commission on Fire Protection. The Operations Level is required of all firefighters for defensively protecting nearby persons, property and the environment from the effects of a harmful release.
- C. Battalion Chiefs will be trained in Hazmat response to the Technician Level.
- D. All program paramedics will be trained in Hazmat response to the Technician Level.

Instructions/Procedures:

I. Initial Operations Response:

- A. Consider the weather at all times.
 - 1. Wind direction is critical during all incidents, especially a hazardous materials incident.
 - 2. Humidity, rain, heat and cold can all play an important role in decision making during this type of incident.
 - 3. It is better to anticipate a wind shift, approaching thunderstorm or even a rise in temperature rather than be surprised by them.
- B. Approach the incident using extreme caution. Resist the urge to rush in until the hazard is known.
- C. Use the three "UPs": Approach upwind, uphill and upstream whenever possible.
- D. Wear full protective clothing with SCBA and secure the scene.
- E. Keep well abreast of the situation and plan an escape route.
- F. Report exact location of the incident to Fire Alarm. Notify incoming support companies of the best entry routes to the scene.
- G. Establish Incident Command:
 - 1. Classify the level of severity of the incident that you expect the incident to progress toward as outlined in II below and, if possible, build responses in progression from Level I to Level IV.
 - 2. Defensively isolate the area as necessary and establish Safety Zones (Hot, Warm and Cold).

3. Identify hazards: Locate and check all placards, container labels, shipping papers and/or knowledgeable sources of information. This action may be too hazardous for close observation; use binoculars.
4. Report leakage, spillage, explosion or fire.
5. Report type and condition of the containment system.
6. Report whether product has entered or is threatening to enter a storm drain, sanitary sewer, or waterway.
7. Assume responsibility for the safety of all personnel involved in the incident.
8. Designate appropriate management and safety positions.
 - a. The IC shall designate a Hazmat Branch Director for Level II, III and IV incidents. The Hazmat Branch Director shall be trained to the Hazardous Materials Technician Level.
 - b. The Hazmat Branch Director shall designate a Hazmat Branch Safety Officer for Level II, III and IV incidents. The Hazmat Branch Safety Officer shall be trained to the Hazardous Materials Technician Level.
 - c. The Hazmat Branch Safety Officer shall be subordinate to the Incident Safety Officer (ISO).

II. Response levels:

A. Level I

1. Description:

- a. A small spill (55 gallons or less) in an open area, or one with an NFPA 704 rating of 2-2-0 or lower.

2. Response:

- a. The battalion satellite company and the closest engine company.
- b. During a Level I, Battalion Chief response will be optional.
- c. Notify the Joint Emergency Operations Center (JEOC) if warranted by Incident Command.

3. Action:

- a. Establish Incident Command.
- b. Refer to the ERG:
 - (1) Find the Guide number.
 - (2) Check Potential Hazards (special attention to those in heavy print).
 - (3) Check Emergency Actions for proper protective clothing and initial actions.

(4) Use other forms of reference to enhance actions

- c. Keep Fire Alarm apprised of the status.
- d. Upgrade the Hazmat Level if conditions warrant.
- e. Remediation will be accomplished in accordance with FWFD SOP S6602.

B. Level II - Established by Level I Incident Command if warranted by the following description:

1. Description:

- a. A small spill or release of materials into the environment that may cause injuries.
- b. Any incident of chemical exposure that will require decontamination.
- c. A large spill of materials (Define large spill), NFPA 2-2-0 or lower.
- d. All fires involving hazardous materials of a significant quantity.
- e. Entry is made into an unknown atmosphere.

2. Response:

- a. Standard one-alarm assignment with a battalion satellite company
- b. B-2, T-2, E-2 and Squad-2
- c. Battalion Chief closest to the incident.
- d. If incident occurs in B-2 the closest a Battalion Chief will respond to assist B-2
- e. TACMED
- f. Med Star
- g. Notify the Shift Commander
- h. Notify Joint Emergency Operations Center (JEOC)
- i. Incident Safety Officer

3. Actions:

- a. Command will be established by the first arriving company and evolve as necessary.

(1) The Command Post will be situated in the Cold Zone, upwind, uphill and upstream, keeping in mind the need for an escape route.

(2) The IC is required to initiate a written Incident Action Plan (IAP) on all Level II or higher incidents. This IAP is started with a simple

organizational chart followed by four ICS forms; ICS 201, 205, 206 and HM 208.

(3) A Planning Section is recommended to assist with ICS forms 201, 205 and 206;

(a) ICS 201 – Incident Briefing

(b) ICS 205 – Communications Plan

(c) ICS 206 – Medical Plan

(4) The Incident Safety Officer and the Hazmat Branch Safety Officer can work together to complete the ICS HM 208 form;

(a) ICS HM 208 – Site Safety and Control Plan

b. At all Level II or higher Hazmat incidents a Hazmat Branch will be established. A Battalion Chief will serve as the Hazmat Branch Director. Hazmat Branch will recommend to Incident Command, any actions that may be necessary in the Hot, Warm and Cold Zone. The Hazmat Branch will designate a Hazmat Branch Safety Officer. The Hazmat Branch Safety Officer will be subordinate to the Incident Safety Officer.

c. Hazmat Branch will establish Safety Zones if not already established by Incident Command. If Safety Zones have been established by Command, Hazmat Branch may change their configuration and locations to meet the needs of the Hazmat team and its operations.

(1) When possible, all zones shall be inside the Incident Perimeter in order to limit public access.

(2) The sizes of these zones will depend on many variables: the physical and chemical makeup of the hazardous materials involved, local terrain, building and population density, weather conditions and hazardous or non-hazardous nature of nearby facilities.

(3) The size of the Warm Zone can be so large that the Incident Perimeter will be too big to control. In such cases, control of intersections and roadways well outside the Warm Zone might be necessary to limit public access.

(4) The ERG can be used to establish initial evacuation distances (Hot Zone) as well as initial downwind protection distances (Warm Zone). These measurements should be used for the first 30 minutes.

(5) These initial distances should be evaluated by the Hazmat Team members and adjusted as necessary.

d. Hazmat Branch will assign Hazmat team duties and responsibilities. These duties and responsibilities shall be carried out using four specific ICS Groups. These four groups are described below. The order of assignment will depend on

the needs of the incident. The exception is the Site Control Group. The Groups are listed below:

- (1) Site Control Group; the duty of the Site Control Group is part of the first arriving company's initial safety measures. When responding to a Hazmat incident, initial arriving companies should query the ERG to control the immediate area. This duty should be continued throughout the incident.
 - (a) This group will operate in the Cold Zone.
 - (b) It is not necessary that the Site Control Supervisor or its members be trained to the Hazmat Technician level.
 - (c) Site Control Group is usually initiated by the initial fire company and eventually can involve more fire companies, police officers and facility personnel.
 - (d) Additional responsibilities of the Site Control Group can include: wind monitoring, placement of incident facilities, downwind protection, barrier tape placement, control of intersections, and protection and enforcement of Safety Zones and Incident Perimeter.
- (2) Research Group; at least two Hazmat Technicians will be utilized to perform research. Research will be performed on behalf of the Hazmat Branch, Hazmat Safety Officer, Decon Supervisor, Entry Supervisor, and Site Control Supervisor. One of the two Hazmat Technicians will be named as Research Supervisor.
- (3) Decontamination (Decon) Group and; the Decon Group Supervisor must be a Hazmat Technician. It is preferred that all members of the Decon Group be Hazmat Technicians. If necessary, Hazmat Operations Level personnel can be utilized.
- (4) Entry Group; the Entry Supervisor and all others entering the Hot Zone are required to be Hazmat Technicians. The Entry Group Supervisor is responsible for tracking air times of all entry team members. Assignments within the Entry Groups are Entry Teams, Backup Teams, Dressing Teams, Air-Time Monitor, and a Medical Monitor.
 - (a) The Entry Team is made up of at least two Hazmat Technicians with appropriate chemical protective clothing (CPC), respiration protection, communications, tools and specific directions from the Entry Team Leader.
 - (b) The Backup Team matches numbers, protection, and capabilities of the Entry Team. There must be a Backup Team in place before entry is made by the Entry Team.

(c) Dressing Teams are used to assist both the Entry Teams and Backup Teams with PPE/CPC donning and doffing, set up of monitors and tools, rehab and any other required assistance. Hazmat Technicians are preferred but not necessary to fill this position.

(d) Air-Time Monitor is a delegated assignment when the Entry Team Supervisor cannot accomplish the task alone.

(e) The Medical Monitor is used to provide medical surveillance for the Entry and Backup Teams. This position is required to be a trained to the Paramedic Level.

C. Level III - Level III Hazmat incidents should be built upon an existing Level II incident if possible.

1. Description:

- a. A release of materials into the environment that has caused numerous injuries.
- b. A release of materials in a congested area with the possibility of injuries or environmental damage.
- c. A release of materials of moderate to high health hazard (Health Hazard of 3 or 4), a moderate to high flammability hazard (Flammability Hazard of 3 or 4), or any reactivity hazard (Reactivity Hazard between 1 and 4). Exception: Confined areas can cause low hazard material to be just as dangerous as those in the moderate to high range.

2. Response:

- a. Level II response plus all available satellite companies.
- b. The Shift Commander and/or Duty Assistant Chief.
- c. Activate the JEOC.

3. Actions:

- a. If available, the Shift Commander may assume Incident Command. However, any Incident Commander at a scene where offensive Hazmat actions are taken must have received appropriate training.
 - (1) In addition to the initial ICS forms required in a Level II response, the IC at a Level III is responsible for three more ICS forms.
 - (a) ICS 202 – Incident Objectives Worksheet
 - (b) ICS 203 – Organization Assignment List
 - (c) ICS 204 – Division Assignment List

(2) It is likely that a Level III response will require extended time to mitigate the hazmat incident. The Incident Commander should identify an operational period and initiate the "Planning P" as referenced by NIMS.

(a) The Planning P is a tool used to forecast and plan for future needs

(b) The Planning Section will be utilized to assist with the next operational period's IAP.

(c) The JEOC can assist with implementation of the Planning P.

b. Hazmat Branch: Ensure Level III actions have been implemented.

c. Site Control Group:

(1) Continue procedures for in-place sheltering or evacuation of citizens as necessary.

(2) Continue to gather information relating to property damage and injuries and determine the need for passenger-carrying vehicles for evacuation.

(3) Monitor the safe route for entry to the Command Post.

(4) Continue to provide incident boundary information to Police Department.

(5) Continue to provide incident perimeter entry information to support agencies.

d. Decon Group:

(1) Continue to operate as in Level II.

(2) Anticipate required rehab, air, and supplies.

(3) Consider the rising level of contaminated runoff and make arrangements for disposal.

e. Entry Group:

(1) Continue to operate as in Level II

(2) Anticipate required rehab, air, and supplies.

(3) Remember, for each entry made, the same number of backup personnel will be needed. Always keep an eye on potential entry personnel and be ready to line up additional Hazmat Technicians to assist in the Backup and Entry teams.

f. Research Group:

(1) Forecast weather that might affect operations during the incident.

(2) Continue to respond to requests for information.

(3) Request additional help when the demands of research prevent timely updates to those requesting information.

D. Level IV - Level IV Hazmat incidents should be built on existing Level III incidents.

1. Description:

- a. A Hazmat emergency of community-wide significance or exhibiting the potential of being a community disaster.
- b. This level incident can cause numerous injuries to the general public and/or damage to the environment.

2. Response:

- a. Level III response
- b. Additional companies as needed
- c. Mutual aid from Arlington or North East Fire Department Association (NEFDA) Hazmat Response Team.

3. Actions:

- a. Ensure Level I, Level II and Level III actions have been implemented.
- b. Implement mutual aid/automatic aid agreements as needed.

III. Fire Alarm:

A. Fire Alarm personnel shall be proficient with and have ready access to the latest ERG.

B. Dispatch Procedures:

- 1. Single Company: The dispatch of a single fire company on a Hazmat emergency will be limited to those incidents of a questionable nature or when a false call is suspected.
- 2. Hazmat Incidents: All other Hazmat emergencies will have a minimum response of a Level I Hazmat incident.
- 3. Alarm will provide the following information:
 - a. Current weather information including: wind direction, velocity, and humidity
 - b. Type and amount of hazardous material
 - c. Type of containment system
 - d. Whether the incident involves a leak, spill, explosion or fire

e. Whether the area has been isolated, evacuated, or has in-place sheltering implemented

f. Information/data from the hazmat file

C. Notification:

1. Hazmat Coordinator, Shift Commander, Duty Assistant Chief, JEOC, and affected staff shall be notified of all Level II or higher Hazmat incidents.

2. City officials as appropriate.

IV. Battalion 2 (if available) shall be responsible for:

A. Assuming Hazmat Branch at all Level II or higher Hazmat incidents.

B. Passing information and action plans to the Incident Commander.

C. Appointing a Hazmat Branch Safety Officer. This Safety Officer will have the authority to halt or suspend all Hazmat Branch operations until safety concerns are addressed.

V. Hazmat Team:

A. The Hazmat Team will be utilized throughout the incident organization and will be trained to the Hazmat Technician Level.

B. The Hazmat Branch shall confer with the Incident Commander or Operations Section to formulate new or different strategies necessary to mitigate the incident. Each Hazmat Group in the ICS organization shall confer with the Hazmat Branch to formulate and coordinate response actions as applicable.

C. The Hazmat Team shall establish or adjust all Safety Zones based on potential danger, taking into consideration materials involved, time of day, wind and weather conditions, monitor readings, incident location, and degree of risk to unprotected personnel. They shall take immediate action to evacuate and/or rescue persons in critical danger, if possible, while providing for the safety of rescuers. Any change in the Safety Zone locations and size must be communicated to Site Control immediately.

D. The Hazmat Team shall identify the materials involved in the situation and the hazards presented by looking for labels, placards, shipping papers, pre-fire plans, and by questioning personnel at the scene (plant management, responsible party, truck drivers). The Team will use reference material carried on their apparatus and will contact CHEMTREC and other agencies, shippers, manufacturers of material, etc. for assistance in sizing up the problem, as necessary.

E. The Hazmat Team shall confer with Hazmat Branch and will use the best information available to formulate an action plan. The properties and danger of the chemical(s) must be realized before the proper selection of personal protective equipment can be made.

F. Hazmat Team members working in the Entry Group, may make an effective entry into the Hot Zone for the purpose of evaluating or mitigating the chemical incident. They will use specialized instruments and equipment to determine the best action plan and will advise the Entry Supervisor of conditions that exist. The Entry Team Supervisor shall be responsible for:

1. Approving personal protective equipment to be worn by the members of the Hazmat Team entering a chemically contaminated atmosphere.
2. Keeping the Hazmat Branch informed of any changes in conditions that might adversely affect the safe operation of the Hazmat Team and other Fire Department personnel.
3. Maintaining communications with other members of the entry team, i.e., by radio/hand signal/visual contact and/or sounding devices.
4. If entry is made into a confined space, FWFD SOP S6403 will apply.

VI. Hazmat Branch Safety Officer Responsibilities:

- A. Ensure the safety of the Hazmat Branch and all personnel involved.
- B. Ensure the following safety measures are implemented:
 1. Entry Supervisor records entry team names and air times
 2. Proper protective equipment and chemical protective clothing is used.
 3. FWFD paramedics are monitoring conditions of each entry team member using entry and exit exams.
 4. Decontamination is properly set up and staffed.
- C. Fill out and maintain the Site Safety and Control Zone Form (ICS 208 HM).

VII. TACMED responsibilities:

- A. Immediately reporting to the Incident Command.
- B. Managing EMS activities on scene, to include:
 1. Assigning other allied health professionals to treat injuries of persons other than responding fire personnel.
 2. Assuring a FWFD paramedic is available for fire personnel working within the safety perimeter.
 3. Utilizing existing protocols and medications in the treatment of chemical injuries according to the authority having jurisdiction.
 4. Assisting Incident Command, Hazmat Branch, and Research Team in researching medical information pertaining to specific hazardous materials.
 5. Disseminating hazardous materials incident information to allied health professionals concerning specific material involved and proper protocol/treatment of injuries sustained based on the best available information.
 6. Performing or supervising pre-entry and exit exams.
 7. Assisting Command with medical debriefing(s).

VIII. Use of Non-Fire Department Personnel:

- A. In some cases it may be advantageous to use non-Fire Department personnel to evaluate hazards and perform certain functions for which they would have particular experience or ability. (Examples: product transfer, manipulation of facility valves, hazmat cleanup, rail car manipulation)
- B. When such personnel are fitted with breathing apparatus, chemical suits, etc., they must be made aware of the functions, limitations and safety precautions necessary in the use of this equipment. The non-fire department worker must have been properly trained and currently proficient in the use of this equipment. Hazmat personnel with the proper personal protective equipment must accompany such personnel for safety purposes.

IX. Decontamination:

- A. The purpose of decontamination is to ensure that any potentially harmful or dangerous residue on persons or equipment is confined within the Hot Zone. Decontamination is intended to prevent the spread of contaminants beyond the Warm Zone.
- B. Specific measures required to decontaminate personnel or equipment will vary with the contaminant, circumstances, and level of contamination. These factors must be considered on an individual basis, within the guidelines described in this procedure.
- C. Hazmat Branch will be responsible for establishing the Decon Group at incidents, which involve a potential contamination problem. As a precaution, Decon Group will be set up for all Level II or higher hazmat incidents.
- D. The Decon Group Supervisor, along with the Research Team, is responsible for determining the most appropriate decon procedures and solutions. The Decon Supervisor is responsible for managing the decontamination process.
- E. The initial assessment of decon requirements must be based upon the specific needs of the situation. This must take into account the specific material involved, degree and type of exposure, and most appropriate decon methods. Assessment will require research and may involve consultation with toxicology resources.
- F. Citizens or emergency response personnel requiring transportation to a medical facility must be decontaminated prior to being transported.
- G. The receiving medical facility shall be notified of the victim's chemical exposure and the proper safety measures to be taken.
- H. The Decon Group Supervisor shall ensure that all personnel and equipment leaving the Warm Zone are not contaminated, and must ensure the following actions:
 - 1. Recon - Verify contamination by the nature of the situation or by the use of instruments.
 - 2. Remove – Perform appropriate decontamination if needed. If no decontamination is needed, record information (see below) and release as necessary.
 - 3. Record – Write down the names of those evaluated and identify, for each:
 - a. The identified contamination material if found

- b. Instruments used or circumstances indicating contamination or no contamination
 - c. The method and solution used to decontaminate the individual
 - 4. Retain – Collect contaminated equipment, articles and decontamination wash solutions. Package them for removal from the site for disposal or decontamination.
 - 5. Release – Release persons after decontamination and evaluation;
 - a. After a medical evaluation, and deemed fit, emergency personnel should be assigned to Rehab, Staging, or an active assignment such as Dressing Team.
 - b. Non-Emergency workers should be medically evaluated and released.
- I. In all cases the primary objective must be to avoid contaminating anyone or anything beyond the Decon Corridor.
- J. In situations, which cause contamination, the Fort Worth Fire Department will use the following procedures to completely decontaminate fire personnel or citizens:
 - 1. Establish an entry point.
 - a. The entry point will be at the edge of the Hot Zone, with the decon corridor spanning the Warm Zone.
 - 2. Complete Decontamination.
 - a. Remove as much of the solid or liquid product as possible and isolate the product.
 - b. Wash off victims while fully clothed. Control the runoff.
 - c. Remove clothing using the available tent and continue to wash the victims.
 - d. Place clothing in bags for cleaning or disposal.
 - e. Depending on the findings of Research and the Decon Supervisor, it is acceptable to vary from all the steps mentioned above. The goal is to remove contaminants. If less invasive steps are acceptable to ensure complete decontamination, some steps can be eliminated. Some examples:
 - 1) Dry Decon – Brushing or blowing off material
 - 2) Emergency Decon – Using water on multiple clothed individuals
 - 3) Vaporization – Allowing a non-persistent vapor contamination to simply vaporize by removing them from the contaminant. (e.g. Carbon Monoxide)
 - 3. When decontaminating emergency personnel, remove and isolate all protective clothing including the SCBA and follow the previous steps.

4. Medical evaluation: All persons involved in the incident, whether citizen or responder, shall be evaluated by a designated FWFD paramedic for vital signs, injuries and symptoms of exposure.

5. All patients previously contaminated shall not be transported by helicopter.

X. Phases of a Hazmat Incident

A. Phase 1: Analyze

1. Collect and interpret hazard and response information.
 - a. Printed and Technical Resources
 - b. Computer Databases
 - c. Monitoring Equipment
 - d. Personal evaluation and eye witness accounts of the situation
2. Consider all scene information as it relates to the incident.
 - a. Weather
 - b. Topography
 - c. Time of Day
 - d. All other contributing circumstances
3. Estimate potential outcomes.

B. Phase 2: Plan

1. Only plan within your capabilities.
2. Identify the incident mode:
 - a. Defensive Mode
 - b. Offensive Mode
 - c. Non-Intervention Mode
3. Identify response objectives from within a specific incident mode.
4. Develop an Incident Action Plan (IAP) and start documenting the response objectives on the following forms:
 - a. An organizational chart
 - b. Incident Briefing (ICS 201)

- c. Communication Plan (ICS 205)
- d. Medical Plan (ICS 206)
- e. Site Safety and Control Plan (ICS 208HM)
- f. During hazmat response Level III or higher, the following forms are required:
 - (a) ICS 202 – Incident Objectives Worksheet
 - (b) ICS 203 – Organization Assignment List
 - (c) ICS 204 – Division Assignment List

C. Phase 3: Implement

- 1. Manage, support and direct resources according to the IAP.
- 2. Assign staff to assist with the IAP formulation.

D. Phase 4: Evaluate

- 1. Evaluate the progress of the planned response.
- 2. Ensure the response objectives are being met safely, effectively and efficiently.
- 3. Adjust the IAP accordingly.
- 4. Evaluation will lead to a new analysis.

E. Phase 5: Terminate

- 1. Transfer of Command to a non-emergency commander (example: clean up contractor)
- 2. Conduct an incident Debriefing.
 - a. As soon as possible after the incident, inform personnel about the chemicals involved and their short- and long-term health effects.
 - b. Schedule medical surveillance for eligible personnel.
 - c. Activate CISD Team as needed.
 - d. The debriefing should cover these key topics:
 - i. Health Information
 - ii. Equipment and Apparatus
 - iii. Exposure Review
 - iv. Follow-up Contact person

- v. Problems Requiring Immediate Action
- vi. Extend gratitude to all who responded

3. Critique the Incident

- a. Three requirements of an effective critique
 - i. Direction – the leader of the critique should lead the meeting with clear objectives and purpose
 - ii. Participation – All vital personnel during the incident should participate
 - iii. Solutions – The goal of the critique should be to find solutions, not to have a fault finding or punitive pretext.
- b. The critique is to identify shortcomings and deficiencies in operations, procedures, training and site plans.
- c. Wait several days before conducting the critique.
 - i. To relieve emotional stress
 - ii. To provide a relaxed and productive atmosphere
 - iii. Allows time to gather relevant information

XI. Evacuation:

A. Factors to consider in recommending evacuation:

1. Explosives or large amounts of material that could explode
2. Large spill of toxic materials
3. Leaking of unknown gases from large containers
4. Leaks that cannot be controlled
5. Vapor density, specific gravity, wind direction, speed, and humidity will play a critical part in this decision.
6. Types of structures, facility and occupancies.
7. The population threatened
8. Weather conditions
9. Time of day

10. The goal of the evacuation is to reduce or prevent public contamination.
11. It must be understood that total evacuation is dangerous, difficult, time consuming, and very expensive. Total evacuation should only be used as a last resort.

B. Evacuation Procedures:

1. Determine the type and extent of material or conditions causing the emergency.
2. Request the activation of the Joint Emergency Operations Center (JEOC.) when outside resources are required to support the evacuation.
3. Obtain the current weather report from the Fire Alarm Office or JEOC and ask to be notified periodically. Also, request immediate notification of any expected changes.
4. Establish a staging area and routes to the area according to the hazard involved.
5. Use maps to determine the area of the evacuation.
6. Divide the area into divisions, utilizing compass directions, i.e., north, northeast, east, etc., or geographical boundaries, street names, etc., for division identification.
7. Notify the Police Department through the Fire Alarm Office of the street intersections to be closed; making sure the area is completely sealed off. Request adequate units be assigned to the area to assure the affected area is isolated. Request police helicopter to evaluate the area to assure all access is sealed off and personnel are evacuated.
8. If feasible, activate the apparatus siren for about 10 seconds, then follow up with an instructional announcement over the public address system.
9. Assign all available Fire Department personnel to expeditiously begin evacuation of the divisions nearest to or in the most danger from the incident. Use Spanish language cards if necessary and available. Make every possible effort to assure that the evacuation message is understood.
10. Assign other departments (Police, etc.) to divisions where the risk of exposure is minimal.
11. Ensure all persons are notified of the need to evacuate. If there are persons who refuse to evacuate in a life or death situation, the Incident Commander must realize that forced evacuation is an extreme measure which could lead to a civil lawsuit against the City.
12. High occupancy buildings (hospitals, schools, office buildings, nursing homes, day care centers, etc.) should have a high priority for notification.
13. As needed, request through the JEOC or the Fire Alarm Office transportation for evacuees to a shelter designated by the American Red Cross or the JEOC.

14. The Incident Commander must be kept aware of the evacuation progress, transportation status, and shelter status.

XII. In-Place Sheltering:

- A. Sheltering In-place is safer, less dangerous, and easier to accomplish than total evacuation. When possible, the IC should consider this option over total evacuation.
- B. Examples of when to Shelter In-Place:
 - a. Sheltering workers inside the business office during a vapor cloud of limited duration
 - b. Moving pedestrians off the street into a building during a hazmat leak
 - c. Moving patients into an uninvolved/safe area of a hospital
 - d. Directing residents back into their homes while a spill is being contained
- C. Citizens shall be directed to:
 1. Go inside their building and remain there until danger passes.
 2. Close all windows and doors.
 3. Shut off all heating, ventilating and cooling systems.
 4. Seal outside doors to the extent possible.
 5. Move to higher levels in high rise structures, if warranted, due to the density of the vapor cloud.

XIII. Reports:

- A. The Station 2 Hazmat on duty Captain is responsible for the proper completion and disposition of the Hazardous Materials Incident Report form F.D. 715. Fill out the F.D. 715 only for incidents of Level II or higher. All hazmat incidents are recorded in the Firehouse Reporting System.
- B. In the event Fort Worth Fire Department personnel are contaminated, a Chemical Exposure Record, Fire Department form F.D. 719, will be completed by the same person completing form F.D. 715. Form F.D. 719 is self-explanatory and distributed from the Hazmat Coordinator's office. **(Must fill out an Occupational Exposure Report DWC-1.)**
- C. File copies of reports shall be retained at Station 2 by the Hazmat Coordinator.
- D. All medical records relating to emergency worker's exposures must be maintained by the Hazmat Coordinator for a total of 30 years after the exposed emergency worker has retired.

References:

Code of Federal Regulations, Title 29 Part 1910.120, Appendix A. United States. U.S. Department of Labor, Occupational Safety & Health Administration (29CFR1910.120)

Fire Fighter's Handbook of Hazardous Materials, Baker, Charles T., 7th edition. (2006). Sudsbury, MA: Jones and Bartlett.

Hazardous Materials: Managing the Incident. Chester Noll, G. G., Hildebrand, M. S., & Yvorra, J. G. (2005). MD: Red Hat Publishing Company, Inc.

NFPA 472: Standard for Professional Competence of Responders to Hazardous Materials Incidents. (2008 ed.). Quincy, MA: NFPA Publications. National Fire Protection Association

Standards Manual for Fire Protection Personnel. Texas Commission on Fire Protection. Austin, TX: Texas Commission on Fire Protection

Hazardous Materials Response Handbook, Fourth Edition, NFPA 2002

IS-0100.b – Introduction to the Incident Command System (ICS), Student Manual

A handwritten signature in black ink, appearing to read 'Ken Stevens', followed by a large checkmark.

KEN STEVENS, SR.
FIRE CHIEF

Sheet Number: _____

Date: _____

IC

SAFETY _____
 PIO _____
 LIAISON _____

ADMIN

PROCUREMENT _____

COST RECOVERY _____

LIAISON _____

LIABILITY _____

RISK MAN. _____

PLANNING

DOCUMENTATION _____

RESTAT _____

SITSTAT _____

TECH. _____
 SPLST. _____

OPERATIONS

STAGING

LOGISTICS

SUPPLIES

COMM EQUIP _____

FUEL/REPAIR _____

FOOD/FLUIDS _____

FIXED FACILITIES _____

DEBRIEFING _____

REHAB _____

HAZMAT

HMB Safety Officer

MEDICAL

DECON

RSRCH

ENTRY

SITE

TRIAGE

TREATMENT

TRANSPORT

All Clear _____
 Undr Cntl _____
 Red Cross _____
 Channels: _____

____ Accountability
 ____ Construction
 ____ Hose Selection
 ____ Communication
 ____ Level II Staging

PAR:

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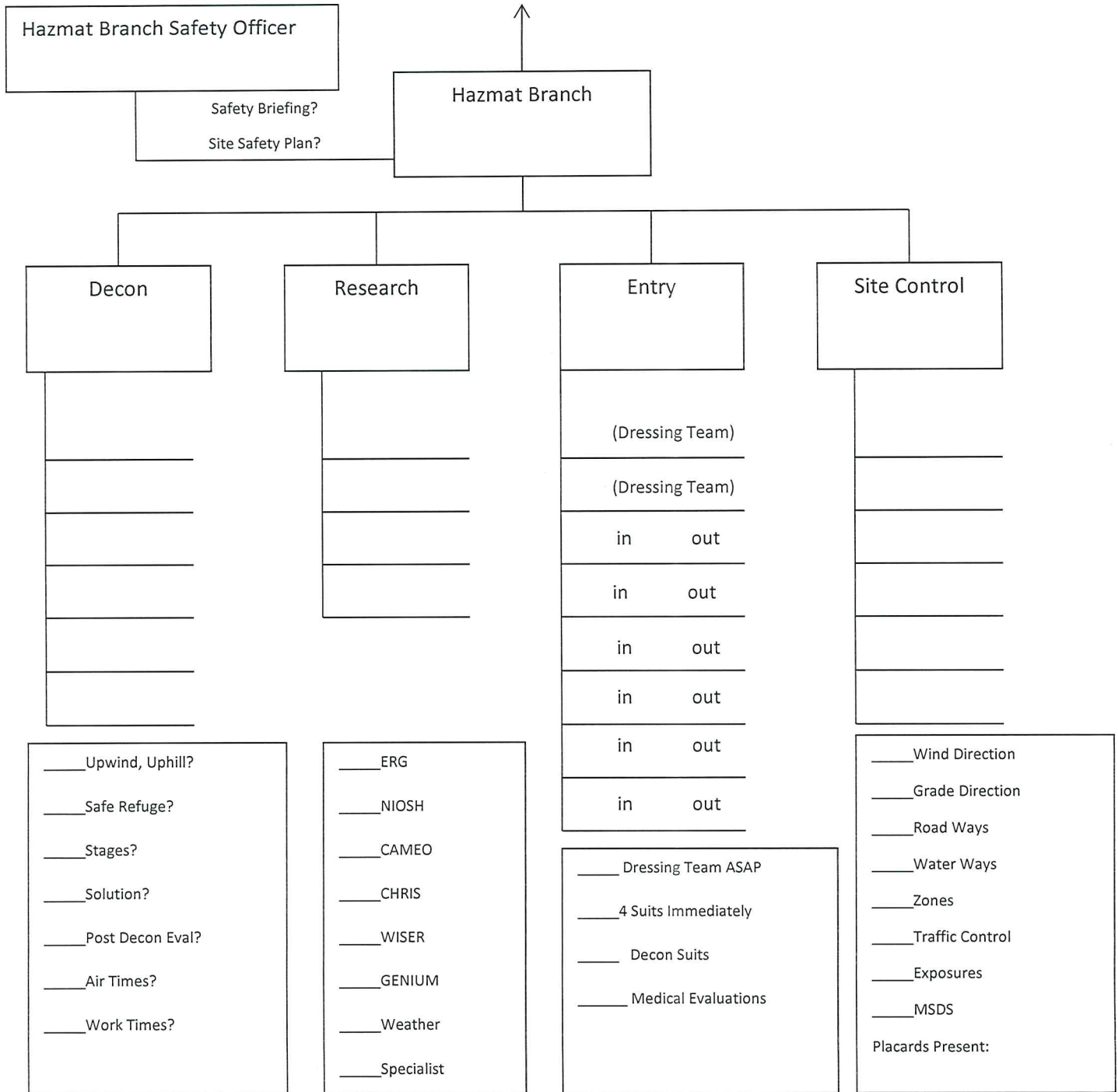
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:60 _____

Staging Area:

Staging Units:



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